

E-Commerce Sales & Customer Behavior Analysis

Exploratory Data Analysis (EDA) Report

Dataset: UCI Online Retail Dataset

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

1. Project Overview

This project analyzes e-commerce transaction data to understand sales performance, product demand, customer purchasing behavior, transaction trends, and potential data issues.

2. Objectives

- Analyze overall sales performance.
- Identify top-selling and frequently ordered products.
- Analyze customer purchasing behavior and identify high-value customers.
- Examine monthly and weekly sales trends.
- Analyze transaction cancellations.
- Identify potential outliers.
- Explore relationships between quantity, price, and sales.
- Generate actionable business insights and recommendations.

3. Dataset & Data Preparation

The dataset contains InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. The original dataset contained 541,909 rows and 8 columns.

Data preparation included removing duplicate rows, removing rows with missing product descriptions, separating cancelled transactions, removing invalid quantity and price values, and creating a Sales column calculated as Quantity × UnitPrice.

After cleaning, the working dataset contained 524,878 rows.

4. Key Business Metrics

Metric	Result
Total Sales	10,642,110.80
Total Quantity Sold	5,572,420
Total Orders	19,960
Unique Customers	4,338
Average Order Value (AOV)	533.17
Cancellation Rate	1.73%

5. Exploratory Analysis

Product Analysis

The analysis examined products by total quantity sold, number of orders, and total sales. WHITE HANGING HEART T-LIGHT HOLDER was the highest by quantity sold and also appeared among the most frequently ordered products. DOTCOM POSTAGE recorded the highest total sales among the listed descriptions, although postage-related descriptions should be interpreted separately from ordinary physical products.

Country Analysis

The United Kingdom generated the highest total sales in the cleaned dataset. Country-level analysis was also performed using transaction counts and average transaction value.

Customer Analysis

The analysis identified 4,338 unique customers. A small group of customers generated particularly high total sales, while another group recorded very high order frequencies.

Sales Trend Analysis

Monthly and weekly sales trends were examined. November 2011 recorded the highest monthly sales (1,503,866.78), while February 2011 recorded the lowest monthly sales (522,545.56) in the cleaned dataset.

6. Visualization Analysis

The notebook includes visualizations covering product and country performance, monthly and weekly sales trends, distributions, box plots, product demand, transaction counts, order frequency, customer sales, scatter plots, and a correlation heatmap. These visualizations were used to identify patterns, relationships, and unusual values in the dataset.

7. Cancellation Analysis

A total of 9,251 cancelled transactions were identified. The cancellation rate was calculated as 1.73% using 9,251 cancelled transactions out of 535,187 transaction rows after duplicate removal.

8. Outlier Analysis

IQR-based analysis identified potential outliers in Quantity, UnitPrice, and Sales. These values were retained because unusually large quantities, prices, or sales may represent legitimate bulk or high-value transactions rather than data errors.

9. Key Insights

- The cleaned dataset generated approximately 10.64 million in sales across 19,960 orders.
- 4,338 unique customers were identified, with an average order value of 533.17.
- November 2011 had the highest monthly sales, while February 2011 had the lowest.
- Several products repeatedly appeared among the highest-selling, highest-quantity, and most frequently ordered products.
- A small group of customers generated particularly high total sales.
- Potential outliers were identified in quantity, unit price, and sales.

10. Business Recommendations

- Focus inventory planning on frequently ordered and high-demand products.
- Monitor high-value customers and encourage repeat purchases.
- Investigate the reasons behind cancelled transactions.
- Use monthly and weekly sales patterns for inventory and promotional planning.
- Review unusual quantities, prices, and sales to distinguish genuine transactions from possible data-entry issues.
- Regularly monitor product performance to identify changes in demand.

11. Conclusion

The exploratory analysis provides a detailed view of e-commerce sales, product demand, customer behavior, and purchasing trends. The findings can support inventory planning, customer retention, and sales decision-making while highlighting areas that may require further investigation.

Note: The reported metrics in this document correspond to the cleaned dataset used in the project.